Masafumi Oizumi, Ph.D.

Associate Professor
Department of General Systems Studies
Graduate School of Arts and Sciences
The University of Tokyo
3-8-1 Komaba, Meguro-ku, Tokyo 153-8902, Japan
Building 16, Room 227B
E-mail: c-oizumi@g.ecc.u-tokyo.ac.jp
URL: https://oizumi-lab-website.vercel.app

EDUCATION

2001-2005 B.S. Department of Physics, The University of Tokyo

2005-2007 M.S. Department of Complexity Science and Engineering, The University of Tokyo (Advisor: Prof Masato Okada)

2007-2010 Ph.D. Department of Complexity Science and Engineering, The University of Tokyo (Advisor: Prof Masato Okada)

WORK HISTORY

Apr, 2019 - present, Associate Professor
The University of Tokyo, Graduate School of Arts and Sciences, Department of General Systems Studies

Mar, 2017 - Mar, 2019, Manager
Araya Inc., Basic Research Group

Sep, 2014 - Oct, 2016, Visiting Researcher
Monash University, Monash Biomedical Imaging, Prof Naotsugu Tsuchiya's Lab

Apr, 2014 - Feb, 2017, RIKEN Special Postdoctoral Researcher
RIKEN, Brain Science Institute, Prof Shun-ichi Amari's Lab

Oct, 2011 - Oct, 2013, Visiting Researcher
University of Wisconsin-Madison, Prof Giulio Tononi's Lab

Apr, 2011 - Mar, 2014, JSPS Superlative Postdoctoral Fellow (SPD)
RIKEN Brain Science Institute, Prof Shun-ichi Amari's Lab

Grants and Fellowships

2021-2031: JST Moonshot R&D

2023-2028: Grant-in-Aid for Transformative Research Areas (A), The Ministry of Education, Culture, Sports, Science and Technology, Japan (MEXT)

2020-2023: Grant-in-Aid for Transformative Research Areas (B), The Ministry of Education, Culture, Sports, Science and Technology, Japan (MEXT)

2018-2024: CREST, Japan Science and Technology Agency (JST)

2018-2022: Grant-in-Aid for Scientific Research (B), The Ministry of Education, Culture, Sports, Science and Technology, Japan (MEXT)

2014-2017: Grant-in-Aid for Young Scientists (B), The Ministry of Education, Culture, Sports, Science and Technology, Japan (MEXT)

2014-2017: RIKEN, Special Postdoctoral Researcher

2011-2014: The Japan Society for the Promotion of Science (JSPS), Research Fellow SPD

2008-2011: The Japan Society for the Promotion of Science (JSPS), Research Fellow DC1

Awards

The First Prize NIKON JOICO AWARD 2021

The University of Tokyo Excellent Young Researcher Award 2019

Asican Pacific Neural Network Assembly Young Researcher Award 2012

Dean Prize, Graduate School of Frontier Sciences (Doctor Course) 2010

Japanese Neural Network Society Research Award 2008

Dean Prize, Graduate School of Frontier Sciences (Master Course) 2007

IEEE CIS Japan Chapter Young Researcher Award 2007

Japanese Neural Network Society Young Researcher Award 2006

Selected Publications

12. Daiki Sekizawa, Sosuke Ito, Masafumi Oizumi (2026)
Koopman Mode Decomposition of Thermodynamic Dissipation in Nonlinear Langevin Dynamics.
Proceedings of the National Academy of Sciences, 123(25), e2530617123. DOI

11. Masafumi Oizumi, Chanseok Lim, Ryota Kanai (2026)
Principal bundle geometry of qualia: Understanding the quality of consciousness from symmetry.
PNAS Nexus, 5(9), pgag261. DOI

10. Daiki Kiyooka*, Ikumi Oomoto*, Jun Kitazono, Yoshihito Saito, Midori Kobayashi, Chie Matsubara, Kenta Kobayashi, Masanori Murayama**, Masafumi Oizumi** (2026)
Single-cell resolution functional networks during unconsciousness are segregated into spatially intermixed modules.
Cell Reports, 45(2), 116902. *, **: equal contribution DOI

9. Genji Kawakita*, Ariel Zeleznikow-Johnston*, Ken Takeda*, Naotsugu Tsuchiya**, Masafumi Oizumi** (2025)
Is my "red" your "red"?: Evaluating structural correspondences between color similarity judgments using unsupervised alignment.
iScience, 28, 3, 112029. *, **: equal contribution DOI

8. Ken Takeda*, Masaru Sasaki*, Kota Abe, Masafumi Oizumi (2025)
Unsupervised Alignment in Neuroscience: Introducing a Toolbox for Gromov-Wasserstein Optimal Transport.
Journal of Neuroscience Methods, 419, 110443. *: equal contribution. The Toolbox is available on GitHub DOI

7. Daiki Sekizawa, Sosuke Ito, Masafumi Oizumi (2024)
Decomposing thermodynamic dissipation of linear Langevin systems via oscillatory modes and its application to neural dynamics.
Physical Review X 14 (4), 041003. DOI

6. Jun Kitazono, Yuma Aoki, Masafumi Oizumi (2023)
Bidirectionally connected cores in a mouse connectome: Towards extracting the brain subnetworks essential for consciousness.
Cerebral Cortex, 33(4), 1383-1402. DOI

5. Shunsuke Kamiya, Genji Kawakita, Shuntaro Sasai, Jun Kitazono, Masafumi Oizumi (2023)
Optimal Control Costs of Brain State Transitions in Linear Stochastic Systems.
The Journal of Neuroscience, 43 (2), 270-281. DOI

4. Masafumi Oizumi, Shun-ichi Amari, Toru Yanagawa, Naotaka Fujii, Naotsugu Tsuchiya (2016)
Measuring Integrated Information from the Decoding Perspective.
PLoS Comput Biol, 12(1), e1004654. DOI

3. Masafumi Oizumi, Naotsugu Tsuchiya, Shun-ichi Amari (2016)
Unified Framework for Information Integration Based on Information Geometry.
Proceedings of the National Academy of Sciences, 113(51), 14817-14822. DOI

2. Masafumi Oizumi*, Larissa Albantakis*, Giulio Tononi (2014)
From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0.
PLoS Computational Biology, 10, e1003588. *: equal contribution DOI

1. Masafumi Oizumi, Toshiyuki Ishii, Kazuya Ishibashi, Toshihiko Hosoya, Masato Okada (2010)
Mismatched decoding in the brain.
The Journal of Neuroscience, 30, 4815-4826. DOI

PUBLICATIONS

Published or in Press

61. Tamami Nakano, Satoe Awada, Masafumi Oizumi, Daisuke Kuwano, Kazuyo Tanji, Nobumasa Kato, Noburo Saji (2026)
Atypical Tactile Expressions Using Japanese Ideophones in Adults With Autism Spectrum Disorders.
Journal of Autism and Developmental Disorders, advance online publication. DOI

60. Ken Takeda, Masafumi Oizumi, Ryo Karakida (2026)
Continual Learning in Modern Hopfield Networks with an Application to Diffusion Models.
Advances in Neural Information Processing Systems (NeurIPS), in press. DOI

59. Mikito Ogino, Daiki Sekizawa, Jun Kitazono, and Masafumi Oizumi (2026)
Designing optimal perturbation inputs for system identification in neuroscience.
eLife, 15, RP110030. DOI

58. Soh Takahashi*, Masaru Sasaki*, Ken Takeda, Masafumi Oizumi (2026)
Investigating Fine- and Coarse-grained Structural Correspondences Between Deep Neural Networks and Human Object Image Similarity Judgments Using Unsupervised Alignment.
Neural Networks, 195, 108222. *: equal contribution. DOI

57. Daiki Sekizawa, Sosuke Ito, Masafumi Oizumi (2026)
Koopman Mode Decomposition of Thermodynamic Dissipation in Nonlinear Langevin Dynamics.
Proceedings of the National Academy of Sciences, 123(25), e2530617123. DOI

56. Ikumi Oomoto, Daiki Kiyooka, Masafumi Oizumi, Masanori Murayama (2026)
Multi-area single-cell calcium imaging dataset of the mouse cortex across wakefulness, sleep, and anesthesia.
Scientific Data, advance online publication. DOI

55. Masafumi Oizumi, Chanseok Lim, Ryota Kanai (2026)
Principal bundle geometry of qualia: Understanding the quality of consciousness from symmetry.
PNAS Nexus, 5(9), pgag261. DOI

54. Yumi Shikauchi, Mitsuaki Takemi, Leo Tomasevic, Jun Kitazono, Hartwig R. Siebner, and Masafumi Oizumi (2026)
Quantifying state-dependent control properties of brain dynamics from perturbation responses.
The Journal of Neuroscience, 46(5), e0364252025. DOI

53. Daiki Kiyooka*, Ikumi Oomoto*, Jun Kitazono, Yoshihito Saito, Midori Kobayashi, Chie Matsubara, Kenta Kobayashi, Masanori Murayama**, Masafumi Oizumi** (2026)
Single-cell resolution functional networks during unconsciousness are segregated into spatially intermixed modules.
Cell Reports, 45(2), 116902. *, **: equal contribution DOI

52. Tomoya Taguchi, Jun Kitazono, Shuntaro Sasai, Masafumi Oizumi (2025)
Association of bidirectional network cores in the brain with perceptual awareness and cognition.
The Journal of Neuroscience, 45(17), e0802242025. DOI

51. Tononi G, Albantakis L, Barbosa L, Boly M, Cirelli C, Comolatti R, Ellia F, Findlay G, Girardi Casali A, Grasso M, Haun AM, Hendren J, Hoel E, Koch C, Maier A, Marshall W, Massimini M, Mayner WGP, Oizumi M, Szczotka J, Tsuchiya N, Zaeemzadeh A (2025)
Consciousness or pseudo-consciousness? A clash of two paradigms.
Nature Neuroscience, 28, 694–702. DOI

50. Tadahiro Taniguchi, Masafumi Oizumi, Noburo Saji, Takato Horii, and Naotsugu Tsuchiya (2025)
Constructive approach to bidirectional influence between qualia structure and language emergence.
Philosophy and the Mind Sciences, 6. DOI

49. Haruka Asanuma, Naoko Koide-Majima, Ken Nakamura, Takato Horii, Shinji Nishimoto, Masafumi Oizumi (2025)
Correspondence of high dimensional emotion structures elicited from video clips between humans and multimodal LLMs.
Scientific Reports, 15, 32175. DOI

48. Asaki Kataoka, Yoshihiro Nagano, and Masafumi Oizumi (2025)
Exploring internal representations of self-supervised networks: few-shot learning abilities and comparison with human semantics and recognition of objects.
Frontiers in Computational Neuroscience, 19, 1613291. DOI

47. Genji Kawakita*, Ariel Zeleznikow-Johnston*, Ken Takeda*, Naotsugu Tsuchiya**, Masafumi Oizumi** (2025)
Is my "red" your "red"?: Evaluating structural correspondences between color similarity judgments using unsupervised alignment.
iScience, 28, 3, 112029. *, **: equal contribution DOI

46. Elise G. Rowe, Ken Takeda, Masafumi Oizumi, Joanita F. D'Souza, Jeroen J. A. van Boxtel, Naotsugu Tsuchiya (2025)
Qualia Structures Collapse for Geometric Shapes, but Not Faces, When Spatial Attention Is Withdrawn.
Neuroscience of Consciousness, 2025(1), niaf053. DOI

45. Asaki Kataoka, Nipun Ravindu Wickramanayaka, Yu Togashi, Masafumi Oizumi (2025)
Spontaneous organization of diverse human color representations: from trichromacy to dichromacy in neural networks trained with retina-like inputs.
In International Conference on Machine Learning and Applications (ICMLA), 1137-1142. DOI

44. Ken Takeda*, Masaru Sasaki*, Kota Abe, Masafumi Oizumi (2025)
Unsupervised Alignment in Neuroscience: Introducing a Toolbox for Gromov-Wasserstein Optimal Transport.
Journal of Neuroscience Methods, 419, 110443. *: equal contribution. The Toolbox is available on GitHub DOI

43. Ken Takeda*, Kota Abe*, Jun Kitazono and Masafumi Oizumi (2025)
Unsupervised alignment reveals structural commonalities and differences in neural representations of natural scenes across individuals and brain areas.
iScience, 28(5), 112427. *: equal contribution. DOI

42. Shunsuke Kamiya, Taiga Mitamura, Muneki Ikeda, Masafumi Oizumi (2025)
Unsupervised Neuronal Matching with Spontaneous Neuronal Activity.
In ICLR 2025 Workshop on Representational Alignment. DOI

41. Daiki Sekizawa, Sosuke Ito, Masafumi Oizumi (2024)
Decomposing thermodynamic dissipation of linear Langevin systems via oscillatory modes and its application to neural dynamics.
Physical Review X 14 (4), 041003. DOI

40. Genji Kawakita, Ariel Zeleznikow-Johnston, Naotsugu Tsuchiya, Masafumi Oizumi (2024)
Gromov–Wasserstein unsupervised alignment reveals structural correspondences between the color similarity structures of humans and large language models.
Scientific Reports 14 (1), 15917. DOI

39. Shunsuke Kamiya, Jun Kitazono, Masafumi Oizumi (2024)
Koopman Operator Based Dynamical Similarity Analysis for Data-driven Quantification of Distance between Dynamics.
In ICLR 2024 Workshop on Representational Alignment. DOI

38. Soh Takahashi, Masaru Sasaki, Ken Takeda, Masafumi Oizumi (2024)
Self-supervised learning facilitates neural representation structures that can be unsupervisedly aligned to human behaviors.
In ICLR 2024 Workshop on Representational Alignment. DOI

37. Jun Kitazono, Yuma Aoki, Masafumi Oizumi (2023)
Bidirectionally connected cores in a mouse connectome: Towards extracting the brain subnetworks essential for consciousness.
Cerebral Cortex, 33(4), 1383-1402. DOI

36. Shunsuke Kamiya, Genji Kawakita, Shuntaro Sasai, Jun Kitazono, Masafumi Oizumi (2023)
Optimal Control Costs of Brain State Transitions in Linear Stochastic Systems.
The Journal of Neuroscience, 43 (2), 270-281. DOI

35. Genji Kawakita, Shunsuke Kamiya, Shuntaro Sasai, Jun Kitazono, Masafumi Oizumi (2022)
Quantifying brain state transition cost via Schrödinger's bridge.
Network Neuroscience, 6 (1), 118–134. DOI

34. Keisuke Ota, Yasuhiro Oisi, Takayuki Suzuki, Muneki Ikeda, Yoshiki Ito, Tsubasa Ito, Hiroyuki Uwamori, Kenta Kobayashi, Midori Kobayashi, Maya Odagawa, Chie Matsubara, Yoshinori Kuroiwa, Masaru Horikoshi, Junya Matsushita, Hiroyuki Hioki, Masamichi Ohkura, Junichi Nakai, Masafumi Oizumi, Atsushi Miyawaki, Toru Aonishi, Takahiro Ode, Masanori Murayama (2021)
Fast, cell-resolution, contiguous-wide two-photon imaging to reveal functional network architectures across multi-modal cortical areas.
Neuron, 109(11), 1810-1824.e9. DOI

33. Kotaro Ishizu, Tomoyo I. Shiramatsu, Rie Hitsuyu, Masafumi Oizumi, Naotsugu Tsuchiya, Hirokazu Takahashi (2021)
Information flow in the rat thalamo-cortical system: spontaneous vs. stimulus-evoked activities.
Scientific Reports, 11(19252). DOI

32. Dror Cohen, Shuntaro Sasai, Naotsugu Tsuchiya, Masafumi Oizumi (2020)
A general spectral decomposition of causal influences applied to integrated information.
The Journal of Neuroscience Methods, 330, 108443. DOI

31. Akihiro Eguchi, Takato Horii, Takayuki Nagai, Ryota Kanai, Masafumi Oizumi (2020)
An Information Theoretic Approach to Reveal the Formation of Shared Representations.
Frontiers in Computational Neuroscience. DOI

30. Jun Kitazono, Ryota Kanai, Masafumi Oizumi (2020)
Efficient search for informational cores in complex systems: Application to brain networks.
Neural Networks, 132, 232-244. DOI

29. Sosuke Ito, Masafumi Oizumi, and Shun-ichi Amari (2020)
Unified framework for the entropy production and the stochastic interaction based on information geometry.
Physical Review Research, 2, 033048. DOI

28. Shun-ichi Amari, Ryo Karakida, Masafumi Oizumi (2019)
Fisher Information and Natural Gradient Learning in Random Deep Networks.
Proceedings of Machine Learning Research, in PMLR 89:694-702. DOI

27. Shun-ichi Amari, Ryo Karakida, Masafumi Oizumi, Marco Cuturi (2019)
Information Geometry for Regularized Optimal Transport and Barycenters of Patterns.
Neural Computation, 31(5), 827-848. DOI

26. Shun-ichi Amari, Ryo Karakida, Masafumi Oizumi (2019)
Statistical Neurodynamics of Deep Networks: Geometry of Signal Spaces.
Nonlinear Theory and Its Applications, IEICE, 10(4), 322-336. DOI

25. Jun Kitazono, Ryota Kanai, Masafumi Oizumi (2018)
Efficient Algorithms for Searching the Minimum Information Partition in Integrated Information Theory.
Entropy, 20(3), 173. DOI

24. Shohei Hidaka, Masafumi Oizumi (2018)
Fast and exact search for the partition with minimal information loss.
PLoS ONE, 13(9), e0201126. DOI

23. Shun-ichi Amari, Ryo Karakida, Masafumi Oizumi (2018)
Information Geometry Connecting Wasserstein Distance and Kullback-Leibler Divergence via the Entropy-Relaxed Transportation Problem.
Information Geometry, 1(1), 1-25. DOI

22. Hiroki Mori, Masafumi Oizumi (2018)
Information integration in a globally coupled chaotic system.
In The 2018 Conference on Artificial Life, 384-385. DOI

21. Andrew M. Haun, Masafumi Oizumi, Christopher K. Kovach, Hiroto Kawasaki, Hiroyuki Oya, Matthew A. Howard, Ralph Adolphs, Naotsugu Tsuchiya (2017)
Conscious Perception as Integrated Information Patterns in Human Electrocorticography.
eNeuro, 4(5), ENEURO.0085-17.2017. DOI

20. Naotsugu Tsuchiya, Andrew Haun, Dror Cohen, Masafumi Oizumi (2016)
Empirical tests of the integrated information theory of consciousness.
In The Return of Consciousness: A New Science on Old Questions, 349-374.

19. Shun-ichi Amari, Naotsugu Tsuchiya, Masafumi Oizumi (2016)
Geometry of information integration.
Information Geometry and its Applications IV (pp. 3-17). Springer, Cham. DOI

18. Masafumi Oizumi, Shun-ichi Amari, Toru Yanagawa, Naotaka Fujii, Naotsugu Tsuchiya (2016)
Measuring Integrated Information from the Decoding Perspective.
PLoS Comput Biol, 12(1), e1004654. DOI

17. Masafumi Oizumi, Naotsugu Tsuchiya, Shun-ichi Amari (2016)
Unified Framework for Information Integration Based on Information Geometry.
Proceedings of the National Academy of Sciences, 113(51), 14817-14822. DOI

16. Melanie Boly, Shuntaro Sasai, Olivia Gosseries, Masafumi Oizumi, Adenauer Casali, Marcello Massimini, Giulio Tononi (2015)
Stimulus Set Meaningfulness and Neurophysiological Differentiation: A Functional Magnetic Resonance Imaging Study.
PLoS ONE, 10(5): e0125337. DOI

15. Masafumi Oizumi*, Larissa Albantakis*, Giulio Tononi (2014)
From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0.
PLoS Computational Biology, 10, e1003588. *: equal contribution DOI

14. Akinori Mitani, Ryo Sasaki, Masafumi Oizumi, Takanori Uka (2013)
A leaky-integrator model as a control mechanism underlying flexible decision making during task switching.
PLoS ONE, 8, e59670. DOI

13. Masafumi Oizumi, Ryota Satoh, Hokto Kazama, Masato Okada (2012)
Functional differences between global pre- and postsynaptic inhibition in the Drosophila olfactory circuit.
Frontiers in Computational Neuroscience, 6:14. DOI

12. Yasuhiko Igarashi, Masafumi Oizumi, Masato Okada (2012)
Theory of correlation in a network with synaptic depression.
Physical Review E 85, 016108. DOI

11. Yosuke Otsubo, Kenji Nagata, Masafumi Oizumi, and Masato Okada (2011)
Influence of synaptic depression on memory storage capacity.
Journal of the Physical Society of Japan, 80, 084004. DOI

10. Masafumi Oizumi, Masato Okada, Shun-ichi Amari (2011)
Information loss associated with imperfect observation and mismatched decoding.
Frontiers in Computational Neuroscience, 5:9. DOI

9. Masafumi Oizumi, Keiji Miura, Masato Okada (2010)
Analytical investigation of the effects of lateral connections on the accuracy of population coding.
Physical Review E, 81, 051905. DOI

8. Yosuke Otsubo, Kenji Nagata, Masafumi Oizumi, Masato Okada (2010)
Instabilities in associative memory model with synaptic depression and switching phenomena among attractors.
Journal of the Physical Society of Japan, 79, 084002. DOI

7. Yasuhiko Igarashi, Masafumi Oizumi, Masato Okada (2010)
Mean field analysis of stochastic neural network models with synaptic depression.
Journal of the Physical Society of Japan, 79, 084001. DOI

6. Ryota Satoh*, Masafumi Oizumi*, Hokto Kazama, Masato Okada (2010)
Mechanisms of maximum information preservation in the Drosophila antennal lobe.
PLoS ONE, 5, e10644, 2010. *: equal contribution DOI

5. Masafumi Oizumi, Toshiyuki Ishii, Kazuya Ishibashi, Toshihiko Hosoya, Masato Okada (2010)
Mismatched decoding in the brain.
The Journal of Neuroscience, 30, 4815-4826. DOI

4. Masafumi Oizumi, Toshiyuki Ishii, Kazuya Ishibashi, Toshihiko Hosoya, Masato Okada (2009)
A general framework for investigating how far the decoding process in the brain can be simplified.
Advances in Neural Information Processing Systems, 21, 1225-1232. DOI

3. Yasuhiko Igarashi, Masafumi Oizumi, Yosuke Otsubo, Kenji Nagata, Masato Okada (2009)
Statistical mechanics of attractor neural network models with synaptic depression.
Journal of Physics Conference Series, 197, 012018. DOI

2. Masafumi Oizumi, Yoichi Miyawaki, Masato Okada (2008)
Rate reduction for associative memory model in Hodgkin-Huxley-type network.
Journal of the Physical Society of Japan, 77(6), 064802. DOI

1. Masafumi Oizumi, Yoichi Miyawaki, Masato Okada (2007)
Higher order effects on rate reduction for networks of Hodgkin-Huxley neurons.
Journal of the Physical Society of Japan, 76(4), 044803. DOI

Preprints

5. Yuria Shimizu, Soh Takahashi, Takato Horii, Masafumi Oizumi (2026)
Relational Knowledge Distillation Brings DNN Representations Close Enough to Humans to Be Aligned Without Supervision.
arXiv:2608.27877. DOI

4. Yu Togashi, Yuko Yotsumoto, Chihiro Hiramatsu, Naotsugu Tsuchiya, Masafumi Oizumi (2026)
Robust individual alignment of color qualia structures: toward a structure-based taxonomy of divergent color experiences.
bioRxiv, 2026.02.13.705699. DOI

3. Ryoichi Watanabe, Jue Wang, Masafumi Oizumi, Noburo Saji, Yusuke Moriguchi (2026)
Unsupervised alignment reveals shared but locally refined colour qualia structure from toddlerhood to adulthood.
bioRxiv, 2026.09.14.751604. DOI

2. Masakazu Inoue, Masafumi Oizumi, Shuntaro Sasai (2025)
Seeing from the inside out: Intrinsic neural dynamics explain individual differences in movie watching.
bioRxiv, 2025.12.03.692223. DOI

1. Nipun Ravindu Wickramanayaka, Masafumi Oizumi (2025)
Systematic comparison of color representations between humans and deep neural networks: towards predicting human color perception in a vast color space.
bioRxiv, 2025.12.10.693611. DOI